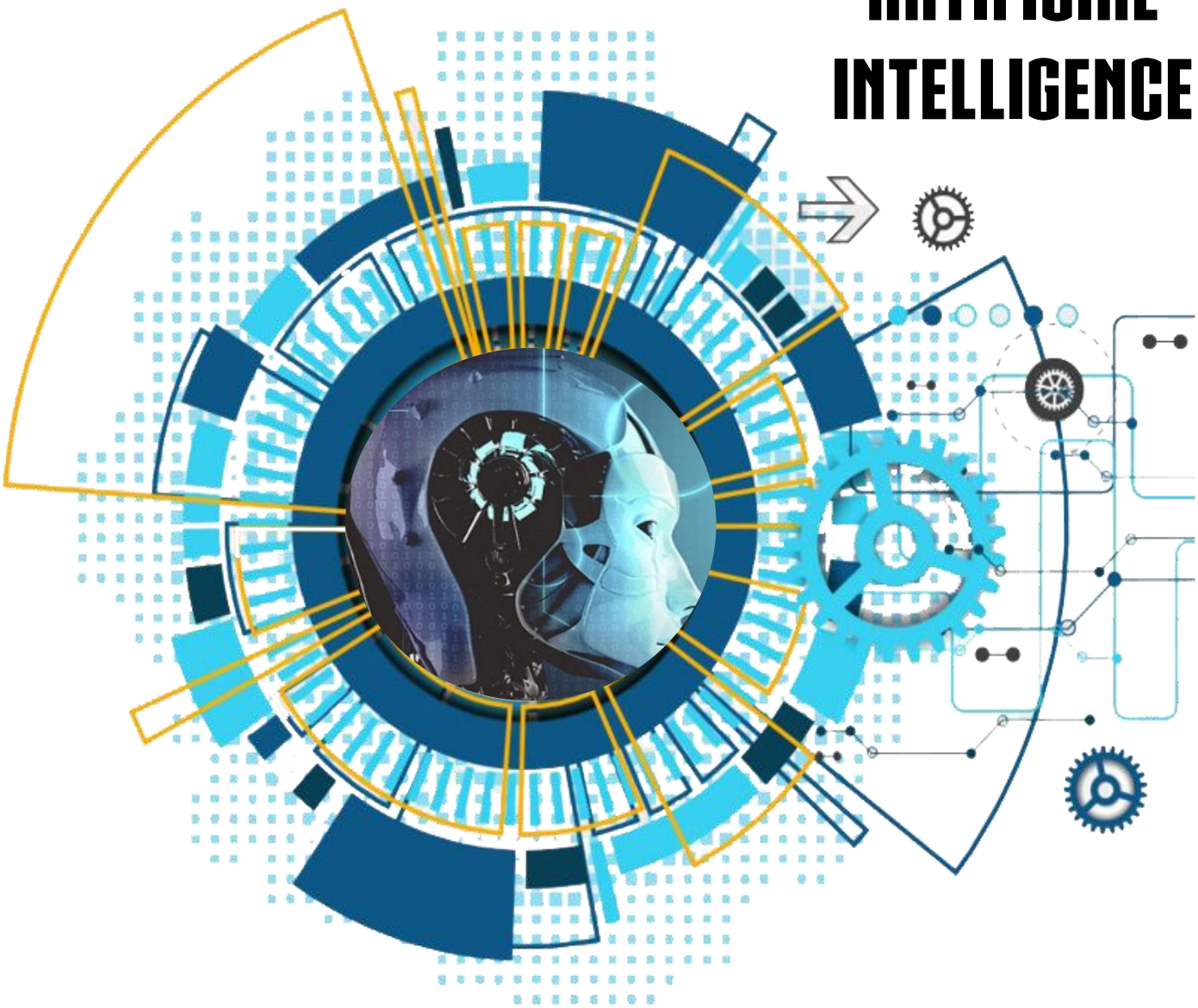
ARTIFICIAL INTELLIGENCE



LEARNING COMPETENCIES

3.1. DETERMINE THE AI AGENTS PROCESS.

3.2. EXPLAIN HOW THE AGENT INTERACTS WITH THE ENVIRONMENT AND HOW THE AGENTS SHOULD ACT IN AI.

ACTIVATING PRIOR KNOWLEDGE

1. What is an Agent?

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   _____

2. What is Artificial Intelligence Agent?

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   _____

   _____

3. What are the three forms of AI Agent?

1.

2.

3

INTRODUCTION

As we know, ideas don't have any value until they are put into action. So, to put artificial intelligence into action, we have an intelligent agent. This agent makes a decision and acts. An AI system consists of two things, first is an intelligent agent, and the second is its environment. An agent requires an environment to carry out its actions.

As human beings, we have hands, ears, eyes, tongue to perceive things and make decisions. In the same way, the intelligent agent will have sensors to perceive the environment. Once we have taken the decision, the next thing is to act upon it. We human beings perform different actions using our expressions, hands, legs, etc. In the same way for AI agent, we have actuators that would perform actions based on a decision made by artificial intelligence.

An intelligent agent (IA) is an entity that makes a decision that enables artificial intelligence to be put into action. It can also be described as a software entity that conducts operations in the place of users or programs after sensing the environment. It uses actuators to initiate action in that environment.

Artificial intelligence is defined as the study of rational agents. A rational agent could be anything that makes decisions, as a person, firm, machine, or software. It acts with the best outcome after considering past and current percepts (agent's perceptual inputs at a given instance). An AI system is composed of an **agent and its environment**. The agents act in their environment. The environment may contain other agents.

An agent is anything that can be viewed as:

- perceiving its environment through sensors and
- acting upon that environment through actuators

Note: Every agent can perceive its own actions (but not always the effects)

DATA INPUT

What is an Agent?

An **agent** is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.

An **agent** can be anything that perceives its environment through sensors and act upon that environment through actuators. An Agent runs in the cycle of **perceiving**, **thinking**, and **acting**.

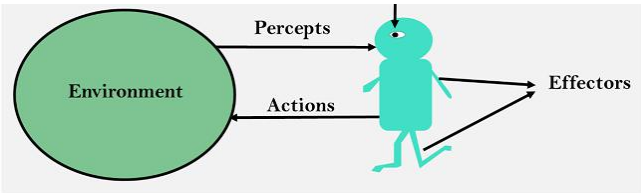
Forms of Agents

- **Human-agent**
  - eyes, ears, skin, taste buds, etc., for *sensors*.
  - hands, fingers, legs, mouth, etc., for *actuators*.
- **Robot agent**
  - camera, infrared, bumper, etc., for *sensors*.
  - grippers, wheels, lights, speakers, etc., for *actuators*.
- **Software Agent**
  - Software agent can have keystrokes, file contents as sensory input and act on those inputs and display output on the screen.

Sensor: Sensor is a device that detects the change in the environment and sends the information to other electronic devices. An agent observes its environment through sensors.

Actuators: Actuators are the component of machines that converts energy into motion. The actuators are only responsible for moving and controlling a system. An actuator can be an electric motor, gears, rails, etc.

Effectors: Effectors are the devices that affect the environment. Effectors can be legs, wheels, arms, fingers, wings, fins, and display screens.



Intelligent Agents

In Artificial Intelligence, an *AI agent* is an acting entity that performs actions to achieve goals, which are set by decisions made using artificial intelligence. Intelligent agents are also called intelligent because they may also learn in the process of achieving goals. In a simple agent, two main functionalities are to percept through sensors and act through actuators. An agent could be very simple as well as very complex; too, it depends on the problem statement.

An *intelligent* agent is an autonomous entity that acts upon an. An intelligent agent may learn from the environment to achieve their goals. A thermostat is an example of an intelligent agent.

Process of Intelligent Agent in AI

An AI agent perceives the environment using its sensors and then with its artificial intelligence makes a decision and via actuators performs actions. Sensors are the medium to provide input to an agent for humans, input sensors are eyes, ears, touch, tongue, etc., in the same way for AI agents it could be cameras, NLP, or output from various sensors. As actuators we humans have hands, legs, expression, and mouth to perform actions, the same way the AI agents have robotic arms, motors, or performing any software integral action. An AI agent is a combination of architecture (the machinery part) and an agent program (functions and conditions).

The following steps are involved in the process of AI agents:

1. An AI agent shall take inputs from the environment using sensors.
2. These observations are then considered for making decisions using artificial intelligence.
3. The decision shall trigger action from agents through actuators.
4. Depending on the agent, it will keep the history of past actions and will use the feedback of past actions in deciding for future events.

Types of AI Agents

Agents can be grouped into five classes based on their degree of perceived intelligence and capability. All these agents can improve their performance and generate better action over the time.

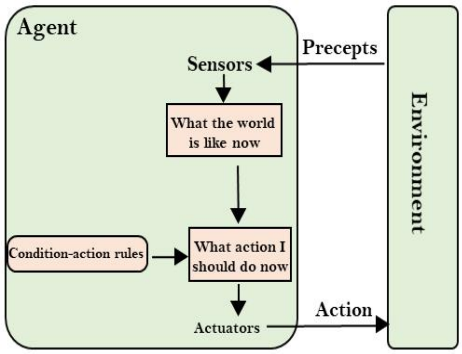
These are given below:

- Simple Reflex Agent
- Model-based reflex agent
- Goal-based agents
- Utility-based agent
- Learning agent

- **Simple Reflex agent:** The Simple reflex agents are the simplest agents. These agents take decisions on the basis of the current percepts and ignore the rest of the percept history. These agents only succeed in the fully observable environment. The Simple reflex agent does not consider any part of percepts history during their decision and action process.

Problems for the simple reflex agent design approach:

- They have very limited intelligence
- They do not have knowledge of non-perceptual parts of the current state
- Mostly too big to generate and to store.
- Not adaptive to changes in the environment.



- **Model-based reflex agent**: The Model-based agent can work in a partially observable environment, and track the situation.

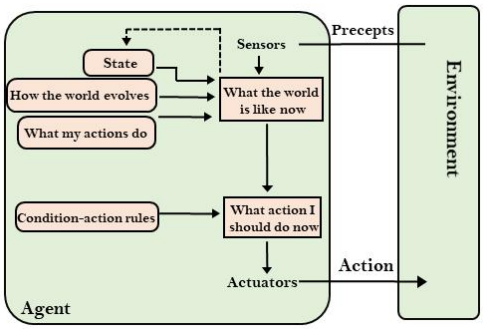
A model-based agent has two important factors:

- **Model:** It is knowledge about "how things happen in the world," so it is called a Model-based agent.
- **Internal State:** It is a representation of the current state based on percept history.

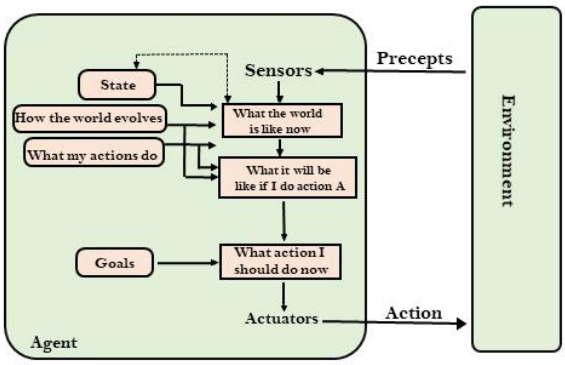
These agents have the model, "which is knowledge of the world" and based on the model they perform actions.

Updating the agent state requires information about:

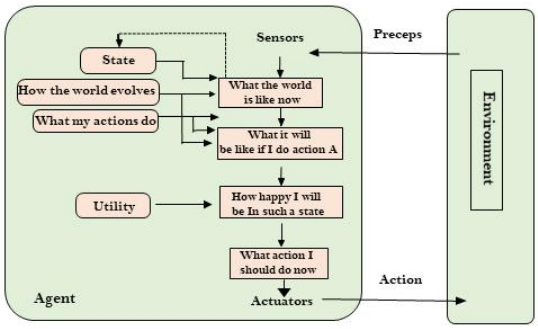
- How the world evolves
- How the agent's action affects the world.



- **Goal-based agents:** The knowledge of the current state environment is not always sufficient to decide for an agent what to do. The agent needs to know its goal which describes desirable situations. Goal-based agents expand the capabilities of the model-based agent by having the "goal" information. They choose action so that they can achieve the goal. These agents may have to consider a long sequence of possible actions before deciding whether the goal is achieved or not. Such considerations of different scenarios are called searching and planning, which makes an agent proactive.



- **Utility-based agents:** These agents are similar to the goal-based agent but provide an extra component of utility measurement which makes them different by providing a measure of success at a given state. Utility-based agents act based not only on goals but also on the best way to achieve the goal. The Utility-based agent is useful when there are multiple possible alternatives, and an agent has to choose in order to perform the best action. The utility function maps each state to a real number to check how efficiently each action achieves the goals.



- **Learning Agents:** A learning agent in AI is the type of agent that can learn from its past experiences, or has learning capabilities. It starts to act with basic knowledge and then can act and adapt automatically through learning.

  A learning agent has mainly four conceptual components, which are:

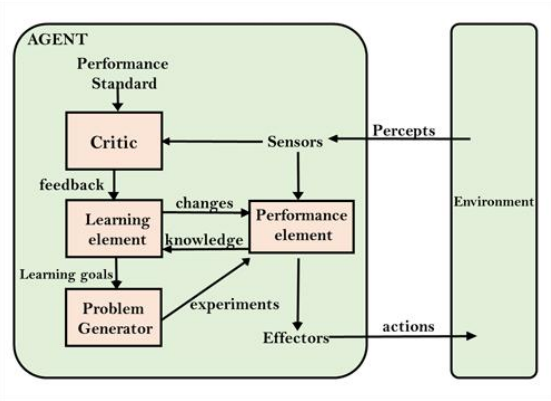
a. Learning element: It is responsible for making improvements by learning from the environment

b. Critic:The learning element takes feedback from the critic which describes how well the agent is doing for a fixed performance standard.

c. Performance element: It is responsible for selecting external action

d. Problem generator: This component is responsible for suggesting actions that will lead to new and informative experiences.

Hence, learning agents are able to learn, analyze performance, and look for new ways to improve the performance



Rules of AI Agent

Following are the main four rules for an AI agent:

- **Rule 1**: An AI agent must have the ability to perceive the environment.
- **Rule 2**: The observation must be used to make decisions.
- **Rule 3**: Decision should result in an action.
- **Rule 4**: The action taken by an AI agent must be a rational.

Benefits of using Intelligent Agent in AI

The following are the benefits of using an intelligent agent.

1. The agent performs actions based upon decisions made by AI. That is to convert ideas into action.
2. An intelligent agent can work on simple commands like a human voice to perform actions, e.g., Alexa, Siri.
3. Intelligent agents do evolve with time, unlike classic agents who can perform a set of predefined actions.

4. AI agents can take utility measurements into account, which makes them more realistic.

The structure of Intelligent Agents

The task of AI is to design an agent program that implements the agent function. The structure of an intelligent agent is a combination of architecture and agent programs. It can be viewed as:

Agent = Architecture + Agent program

The IA structure consists of three main parts: architecture, agent function, and agent program.

1. **Architecture:** This refers to machinery or devices that consist of actuators and sensors. The intelligent agent executes this machinery. Examples include a personal computer, a car, or a camera.
2. **Agent function:** This is a function in which actions are mapped from a certain percept sequence. Percept sequence refers to a history of what the intelligent agent has perceived. It is produced through the agent program's execution on the physical architecture.
3. **Agent program:** This is an implementation or execution of the agent function.

The agent function PEAS Representation

PEAS is a type of model on which an AI agent works upon. When we define an AI agent or rational agent, then we can group its properties under the PEAS representation model. It is made up of four words:

- **P:** Performance measure
- **E:** Environment
- **A:** Actuators
- **S:** Sensors

Here performance measure is the objective for the success of an agent's behavior.

PEAS for Self-Driving Car

Let's suppose a self-driving car then PEAS representation will be:

Performance: Safety, time, legal drive, comfort
Environment: Roads, other vehicles, road signs, pedestrian
Actuators: Steering, accelerator, brake, signal, horn
Sensors: Camera, GPS, speedometer, odometer, accelerometer, sonar.